Interoperability Specifications for Communication with Systems field

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1. General concepts

1.1. Introduction

The standards covered impact the following operating contexts:

- **Device Management:** management of the **operational configurations of the devices** that they make up the physical monitoring infrastructure;
- **Data Flow Management:** processes for **sharing and processing** recorded and/or processed locally by field systems towards the IoT Central Platform;
- **Alert Management:** notification of **faults** or **alarms** to the IoT Central Platform, in the event of threshold exceedance, malfunctions or changes in network status.

1.2.2. Context And Hiring

The reference context in question includes defining the integration solution between RFI's IoT Platform and monitoring systems. The following is the logical architecture representing the bridge monitoring systems.

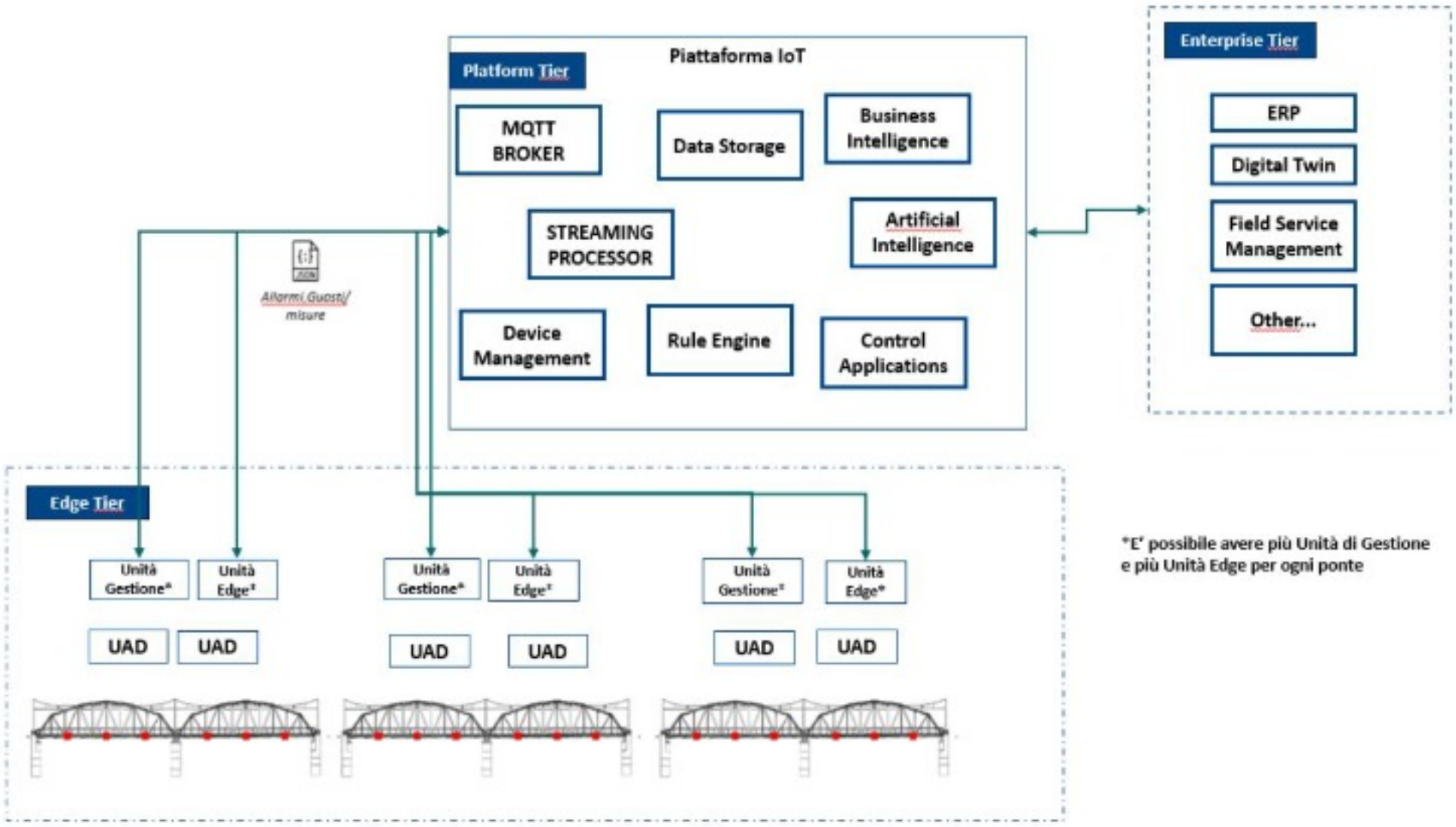

Figure 2 - Logical Architecture

As shown in Figure 2 – Logical Architecture, the data detected by the sensors installed on the bridges are collected by the Data Acquisition Units (DAU) to which they are directly connected.

The field architecture includes at least one UAD for each instrumented span, capable of collecting information from the various sensors installed on the particular monitored span.

Each UAD sends data to a Management Unit. Each Management Unit sends data to one or more Edge units, consisting of an industrial PC, positioned inside an outdoor cabinet near the instrumented structure. Depending on the number of spans, each bridge can have multiple Management Units and Edge Units.

The developments on the Edge Unit and the Management Unit will be entirely borne by RFI and the respective suppliers according to the tender specifications and to date no product installations by FS Technology are foreseen on these components.

The IoT platform provides advanced capabilities including data streaming, historicization, and analytics.

The data acquired in the IoT platform will be used for various purposes of the maintenance process:

- Alarm/fault management; •
  Maintenance process optimization; • Acquisition
  and logging of measurements for timely analysis of the state of the artwork.

The logical architecture shown (Fig. 2) provides for a two-way data exchange between the field units (Management and Edge) and the IoT Platform. In fact, there will also be a dedicated *flow* to the Management Units or Edge Units:

- to the transmission of status changes made at central level of faults and alarms respectively;
- to the transmission of commands for the management of sensor devices.

1.2.3. Objectives of the Specification

The objective of the specification is to define communication methods for integrating the various types and families of data between the Management and Edge Units and RFI's IoT platform. The objectives of the specification are divided into three macro-functional contexts:

- Alert Management - involves the detection, filtering and notification of alarm events or failures towards the IoT Platform;
- Data Flow Management - concerns the processes and protocols for acquiring and transferring field-measured quantities to the centralized data storage, processing, and visualization systems of RFI's IoT platform. This also includes the data produced as output by the processing algorithms residing at the Edge (e.g., structure frequencies, performance indicators, etc.).

- Device Management - concerns the management of the status and operational configurations of the devices that make up the physical monitoring infrastructure such as sensors, acquisition systems and field processors (e.g. Management Units, Edge Units, UADs, Dataloggers, Local Data Storage Units, etc.)

The specifications in this document will be illustrative in nature, with a good level of detail, but they will not be completely exhaustive as they will be detailed during the implementation phase. Therefore, any changes are subject to change. Such changes will be communicated prior to the start of field systems development and will not affect the nature and *effort* expected for the development of the field systems contractors.

2. Communication Modes Field Systems Unit – IoT Platform

2.1 Authentication of Management and Edge Units

Authentication will involve the use of a TLS digital certificate (X.509) by the Management Unit/Edge Unit to communicate information to the IoT platform.

The following differentiation is expected depending on the protocol used:

Protocol	Authentication
MQTTS User	+ password
SFTP	Username + password
HTTPS	JWT Token

Additionally, an Access Control List is provided to manage data sending authorization for a particular sensor/UAD/ Management Unit/Edge Unit/etc.

2.1.1 Authentication with JWT Token

In order for the Management Unit/Edge Unit and the IoT platform to exchange data, in the case of HTTPS communication, it is necessary to include the JWT authentication token in the HTTP header.

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The process of creating the JWT token is as follows:

- Token Request: Before making an API call, the device requires a JWT token to the Identity Provider (IdP)
- If the authentication is valid, the authentication server returns a JWT token to the caller
- Token Usage for REST API Calls: Once obtained, the JWT token is included in the HTTP Authorization header of each request. Specifically, the token is passed in the HTTP Authorization header of each request: e.g., GET https://{URL}/api/devices/status Authorization: Bearer eyJhbGciOiJIUzI1NiIs....

2.2 Specification Summary

The following table provides, based on the type of flows, a summary of the protocols and data formats used in the messages exchanged between the Management Unit/Edge Unit and the IoT Platform.

During the implementation phase, detailed implementation cases will be evaluated.

The table below defines the Quality of Service levels of the MQTT protocol, but this parameter must be configurable and therefore must be modifiable during the implementation phase without any implementation impact.

Flow	Case Studies of Use	Protocol	Data Format
Alert Management - Alarm Management	Near-Real Time Alarm Sending	MQTTS QoS=1	json
Alert Management - Fault Management	Near-Real Time Fault Reporting	MQTTS QoS=1	json
Data Flow Management - Measurements and Results Algorithms	Send measurement/result at reduced frequency and small payload size (KB order)	MQTTS QoS=0	json
	Massive sending of measurement/ result with medium and large payloads	SFTP; SFTP+HTTPS	CSV + ZIP
Data Flow Management - Snapshot	Sending small images in Base64 and at a reduced frequency	MQTTS QoS=0	json
	Bulk sending of medium and large sized images	SFTP; SFTP+HTTPS	ZIP Images
Device Management	Sending commands	MQTT QoS=1	json

The document defines the data exchange methods, but the definition of field domains (i.e., master data) and the related valuation logic are not covered by these specifications and will be defined during the implementation phase between RFI and the field systems supplier. Furthermore, these master data must be communicated to FS Technology accordingly.

The frequency of data transmission to the IoT Platform will be determined on a case-by-case basis between RFI and the supplier. *FS Technology will need to be involved to assess the impact on the IoT Platform.*

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2.3 Protocols

2.3.1 MQTTS Protocol

The MQTTS protocol supports real-time and/or near-real-time communication between the Management Unit/Edge Unit and the IoT platform in both directions. It is a lightweight messaging protocol designed for applications requiring low data impact and/or limited bandwidth. Adopting this protocol therefore aims to standardize communication with various vendors' field systems, as well as ensure data stability and persistence.

In particular, the MQTTS protocol is a standard ISO protocol capable of guaranteeing:

- Homogeneous standard protocol;
- Data stability and persistence; •
  Continuous and unsolicited mode (Push); •
  Bidirectional asynchronous transmission (from the field to the platform and vice versa); • Low impact on the network and processing resources.

Details of the selected protocol are below:

- MQTT 5.0 (release date 2019) defines a TCP/IP data transmission protocol based on a publish/ subscribe model that operates through a dedicated message broker located in the IoT Platform. If a later version is used, it must be backwards compatible with version 5.0.

- TLS (Transport Layer Security) is used to encrypt communication between the client and the MQTT broker.
- The port used by the MQTT protocol must be configurable and will be specified during the implementation phase.

- The Quality of Service can be of type 1 or 0 depending on the flow, and must be a configurable parameter and therefore must be modifiable during the implementation phase without any implementation impact.

2.3.2 SFTP Protocol

The use of the SFTP protocol is foreseen in the case of batch transmission of data from the Unit of Edge Management/Unit to IoT Platform

SFTP (Secure File Transfer Protocol) is a network protocol designed for secure file transfer over an encrypted connection. Based on the SSH (Secure Shell) protocol, SFTP provides strong authentication and data protection during transmission, preventing interception and tampering.

SFTP is widely used to ensure the integrity and confidentiality of sensitive information, such as operational data, reducing the risk of unauthorized access and man-in-the-middle attacks.

2.3.3 HTTPS Protocol

This protocol is expected to be used in case of a request via API by the IoT Platform to re-forward data in the event of a fault or to send implementation commands to the field component (Device Management). The HTTPS (Hypertext Transfer Protocol Secure) protocol is a communication protocol used for the secure transmission of data over the Internet. It is based on the HTTP (Hypertext Transfer Protocol), but adds a layer of encryption and authentication to ensure the

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Privacy and security of data exchanged between the client and the server. HTTPS is widely used to protect the confidentiality of sensitive information.

3. Alert Management and Data Flow

The information transmitted by the monitoring system to the platform has been classified into the following **data catalogue** (alarms, measurements, faults):

1) Alarms: These indicate the abnormality of a specific parameter and provide a logical True/False status, complemented by indications of the alarm's increasing intensity. These are further classified into:

- **Pre-alert (Events):** Refers to the pre-alert level and with increasing intensity.
- **Alert (Alarm):** Relating to the abnormal state of a specific parameter, defined by a logical True/False state and completed by indications for the alarm level, with increasing intensity.

The distinction between messages containing event data and those related to alarms is managed through the "AlarmSeverityLevel" field in the JSON, described below. The classification will be established by RFI when defining the master data.

2) **Faults** (or failures): Information regarding the identification and reporting of abnormal operation and/or failure of one or more components of the monitoring system (sensor, acquisition unit, management unit, edge unit, switch, router, etc.). This information can be correlated and also provides support for device management and device management in general.
3) **Measurements:** Quantitative information of the measurement acquired by the sensors. This flow can be also used for sending algorithm results.
4) **Snapshot:** Images acquired on the monitored structure.

3.1 Alarm Management

As regards the "Alarm Management", the messages to be managed from field systems to the platform IoT are:

- Alarm onset • Alarm reset

While those from IoT platform to the Management Unit:

- Alarm acknowledgement
- Alarm closure

Below is a description of the JSON fields for the aforementioned messages. During implementation, FS reserves the right to increase the number of JSON fields by up to 50% and update the description and message type shown below without incurring any additional charges from the Contractor.

Field	Description	Message Type
MsgId	Unique identifier of the sent message that allows the differentiation of all received messages	Alarm onset Alarm acknowledgement Alarm reset Alarm closure

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IDGateway	Unique identifier of the Management Unit or Edge Unit that sends the information, allows you to quickly trace the affected device	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
SenderId	Unique supplier identifier of the	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
Timestamp	Timestamp of the alarm (timestamp when the problem actually occurred). Format: Milliseconds since Unix Epoch	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
Version	Software version	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
SensorAlarm{AlarmId, Value}	Where AlarmId : Unique identifier of the alarm given the asset Value: Value assumed by the measuring device: "1" for opening	Alarm onset
SensorAlarmClosed{AlarmId, Value}	Where AlarmId : Unique identifier of the alarm given the asset Value: Value assumed by the measuring device: generally "0" for return	Alarm reset
SensorAlarmHandled{AlarmId, Value}	Where AlarmId : Unique identifier of the alarm given the asset Value: Value assumed by the measuring device, "2" for acknowledgement	Alarm acknowledgement
SensorAlarmManagedClosed{AlarmId, Value}	Where AlarmId : Unique identifier of the alarm given the asset Value: Value assumed by the measuring device: "3" for manual closing	Alarm closure
CurrentValue	Current value (above the threshold) of the individual alarm. This allows you to provide information on how much has been exceeded and perform subsequent processing.	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
Area	Sector that identifies the type of alarm (LV, TLC, TE, IS)	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
FaultTimestamp	Alarm timestamp (timestamp when the message was sent to IoT). Format: Milliseconds since Unix Epoch	Alarm onset
Problem Intensity	Text field where the intensity description is entered. This field must be mapped in the master data.	Alarm onset Alarm acknowledgement Alarm reset Alarm closure

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Problem Location	A text field describing the location of the received alarm, providing additional information to the Latitude and Longitude fields. This field must be mapped in the registry.	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
AlarmText	Free text field where you can add additional information if necessary	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
Alarm Severity Level	Identifier of the severity level of the alarm generated	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
SensorList [{IdUAD, IdSensor, SensorType }]	list of sensors that generated the accompanying alarm (minimum 1) IdUAD: Identifier of the Sensor afferent UAD IdSensor: Unique identifier of the sensor that generated the alarm SensorType: Identifier of the type of belonging of the sensor	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
SpanId	SAP identifier of the span of the work affected by the alarm.	Alarm onset Alarm acknowledgement Alarm reset Alarm closure
Pkalarm	Unique alarm identifier. Generated by the field system to identify repeated identical alarms (same pkalarm) from different sequential alarms (different pkalarm).	Alarm onset Alarm acknowledgement Alarm reset Alarm closure

Table 1 Application/JSON for alarm management

Alarms with the same primary key (Pkalarm) must not be sent back to the IoT Platform unless a realignment operation is explicitly requested. However, they must be sent if they undergo a status change, such as a change from "Alarm Onset" to "Alarm Closure," or if their severity changes.

Additionally, in the event of an alarm affecting multiple spans, the field systems must send an alarm message for each affected span. Each message will have the same pkAlarm and will contain the sensors affected by the alarm (via the SensorList array) that pertain to the specific span (spanId).

3.1.1 Periodic Alarm Realignment

A configurable flow from the field to the IoT platform for periodic alarm status realignment will also be required, with the same characteristics as the flows described above. The details will be defined during the implementation phase.